

Scientific Computation

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Lecture 16 - Linear Regression

Convergence Theorems

The Jacobi and Gauss-Seidel methods have the following convergence results (which we will not prove):

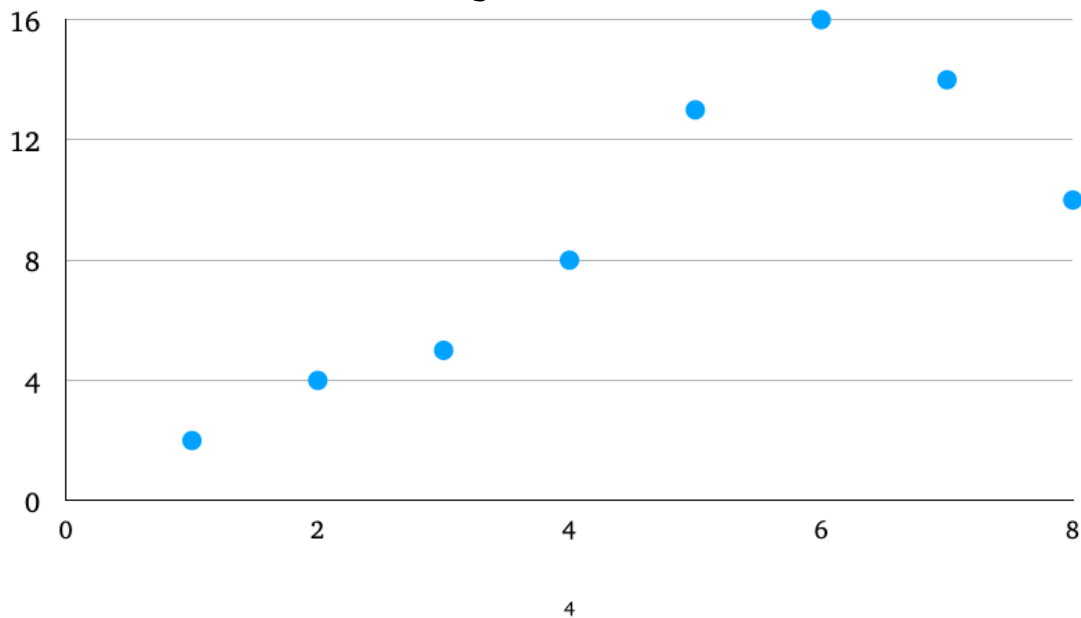
Lecture Outline

- **Curve fitting**
- **Linear regression**
- **Linear regression in matrix form**

Curve Fitting

Given a set of points x_i and observations y_i , **curve fitting** or **smoothing** tries to fit a function that captures the association between x and y for other points:

$$v(x_i) \approx y(x_i)$$



Curve Fitting Formulation

Consider a set of **basis functions** ϕ_i , $i = 1, 2, \dots, m$, where $m \neq n$ in general. Our problem is to find

$$y = v(x) = \sum_{j=1}^m c_j \phi_j(x)$$

*c is scalar weights
we are trying to find
(parameters)*

such that the magnitude of **residuals**

$$r_i = v(x_i) - y_i, \quad i = 0, \dots, n$$

is minimized. *It's the square of the Euclidean distance, and it penalizes large deviations.*

least squares is related to the best estimation

stats: least squares is related to max likelihood estimator for Gaussian noise.

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Optimization Problem

We will assume that minimizing the residuals is in a **least-squares sense**, i.e., our problem is to solve

$$\min_{c \in \mathbb{R}^m} \sum_{i=1}^n (v(x_i) - y_i)^2$$

where $v(x_i) = \sum_{j=1}^m c_j \phi_j(x_i)$.

Why have we chosen to minimize squared residuals $r_i^2 = (v(x_i) - y_i)^2$?
A few reasons:

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The Simplest Case: Linear Regression

Let's start by fitting $n + 1$ points (x_k, y_k) with a linear function:

$$y = ax + b,$$

i.e., $\phi_1(x) = b$, $\phi_2(x) = ax$, and $c = [a \ b]^\top$. This leads to the optimization problem

$$\min_{a,b \in \mathbb{R}} \phi(a, b)$$

where $\phi(a, b) = \sum_{i=0}^n \underbrace{(ax_i + b - y_i)^2}_{V(x_i)} = \sum_{i=0}^n r_i^2.$

First-Order Optimality for Linear Regression

Regression

This is an unconstrained optimization problem in two variables, so our necessary first-order optimality conditions are

$$\frac{\partial \phi}{\partial a} = 0, \frac{\partial \phi}{\partial b} = 0.$$

$$0 = \frac{\partial \phi}{\partial a} = 2 \sum_{i=0}^n (ax_i + b - y_i) x_i \quad \text{chain rule}$$

$$\Rightarrow a \sum_{i=0}^n x_i^2 + b \sum_{i=0}^n x_i - \sum_{i=0}^n y_i x_i = 0$$

$$\underbrace{\left(\sum_i x_i^2 \right) a + \left(\sum_i x_i \right) b}_{\text{linear in } a \text{ and } b} = \sum_i y_i x_i$$

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First-Order Optimality Cont.

Now let's examine $\frac{\partial \phi}{\partial b} = 0$:

$$0 = \frac{\partial \phi}{\partial b} = 2 \sum_{i=0}^n (ax_i + b - y_i)$$

$$\Rightarrow a \sum_i x_i + b \sum_i 1 - \sum_i y_i = 0$$

$$\left(\sum_i x_i \right) a + (n+1)b = \sum_i y_i$$

A Linear System for Linear Regression

We can combine these to get a linear system:

$$\begin{bmatrix} \sum_i x_i^2 & \sum_i x_i \\ \sum_i x_i & n+1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \sum_i x_i y_i \\ \sum_i y_i \end{bmatrix}$$

$$A \in \mathbb{R}^{2 \times 2}$$

$$\in \mathbb{S}_+^{2 \times 2}$$

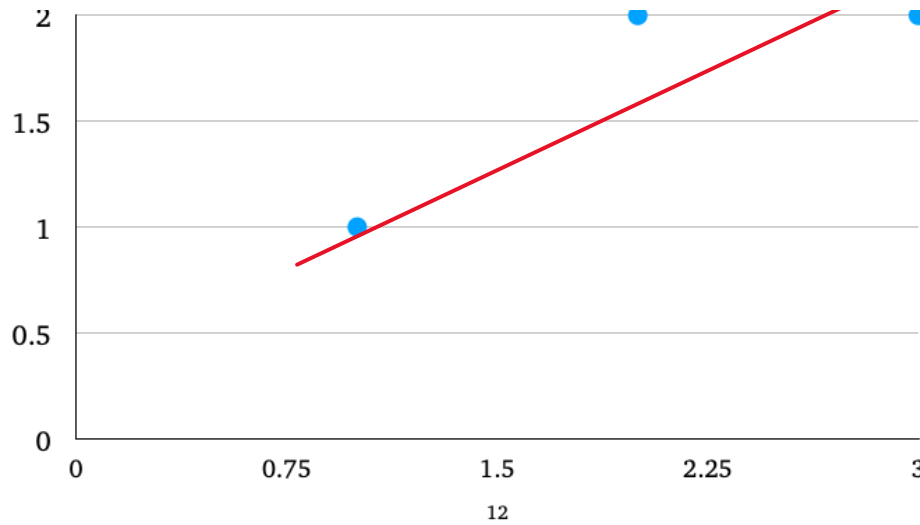
coeff are
solved

Linear Regression Example

Example 1: Consider the dataset

	0	1	2	n
x_i :	1	2	3	
y_i :	1	2	2	

visualized below. What is the best linear fit $y = ax + b$?



Linear Regression Example

Example 1 cont.: To find a and b , we solve

$$\begin{bmatrix} \sum_{i=0}^n x_i^2 & \sum_{i=0}^n x_i \\ \sum_{i=0}^n x_i & n+1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \sum_{i=0}^n x_i y_i \\ \sum_{i=0}^n y_i \end{bmatrix}$$

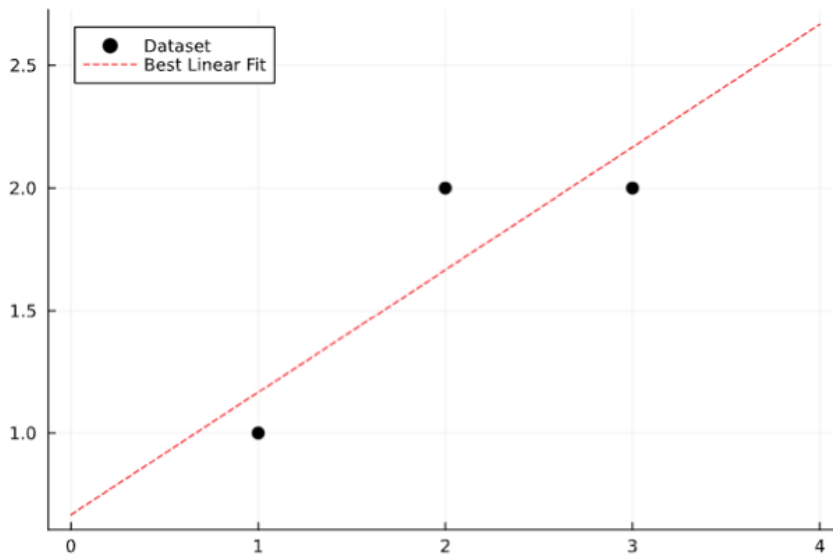
$$\begin{bmatrix} 1^2 + 2^2 + 3^2 & 1 + 2 + 3 \\ 1 + 2 + 3 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 + 4 + 6 \\ 1 + 2 + 2 \end{bmatrix}$$

$\downarrow$ $\downarrow$
 $\begin{bmatrix} 14 & 6 \\ 6 & 3 \end{bmatrix}$ $\begin{bmatrix} 11 \\ 5 \end{bmatrix}$

Linear Regression Results

Linear Regression Results

Example 1 cont.: In Julia, we get the following fit:



```
using Plots
x = [1; 2; 3]
y = [1; 2; 2]
n = length(x)
A = [sum(x.^2) sum(x)
      sum(x)    n]
b = [sum(x.*y); sum(y)]
display(A)
c = A\b
x_plot = 0:0.01:4.0
linear_fit(x) = c[2] + c[1]*x
scatter(x, y, label="Dataset",
        markersize=5, color=:black)
plot!(x_plot, linear_fit.(x_plot),
       label="Best Linear Fit",
       color=:red, linestyle=:dash)
```

✓ 4.0s

2x2 Matrix{Int64}:

14	6
6	3

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Linear Regression Once More

Let's rewrite our residuals for linear regression as a linear system:

$$r_i = v(x_i) - y_i = ax_i + b - y_i$$

$$r = \begin{bmatrix} ax_0 + b \\ ax_1 + b \\ \vdots \\ ax_n + b \end{bmatrix} - y = \begin{bmatrix} x_0 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} - y$$

$\phi_2 \quad \phi_1$

Each row is an equation induced by x_i, y_i

*Each column corresponds to one basis function
in our model $v(x)$*

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An Overdetermined System

We have the linear system

$$0 = r = \begin{bmatrix} x_0 & 1 \\ x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} - y \in \mathbb{R}^{n+1},$$

$A \quad z$

L " J

which has $n + 1$ equations but only 2 unknowns, so it is **overdetermined**.
Note the “tall” shape of the matrix.

$$v = r = Az - y$$

$$\Rightarrow \text{so we minimize } \|r\|_2^2 = \sum_{i=0}^n r_i^2$$

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Matrix Form of Least Squares

We can now write the whole procedure in the following abstract form:

$$\min_{z \in \mathbb{R}^m} \|Az - y\|_2^2 \quad \text{in } 2 \text{ norm}$$

$$\begin{aligned} \|Az - y\|_2^2 &= (Az - y)^T (Az - y) \\ &= z^T A^T A z - \underline{z^T A^T y} - y^T A z + y^T y \\ &= z^T A^T A z - 2 z^T A^T y + \|y\|_2^2 \end{aligned}$$

1x1

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Matrix Calculus

MATRIX CALCULUS

We can optimize this with some matrix calculus:

$$f(z) = \|Az - y\|_2^2 = z^T A^T A z - 2 z^T A^T y + \|y\|_2^2$$

Unconstrained minimization: take $\nabla f(z) = \begin{bmatrix} \frac{\partial f}{\partial z_1} \\ \vdots \\ \frac{\partial f}{\partial z_n} \end{bmatrix} = \vec{0}$

$$0 = \nabla f(z) = 2A^T A z - 2A^T y$$

$$\Rightarrow A^T A z = A^T y$$

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Comparing with Prior Approach

Let's expand this solution to check whether it matches our prior approach:

$$\begin{aligned} A = \begin{bmatrix} x_0 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} & \Rightarrow A^T A = \begin{bmatrix} x_0 & \dots & x_n \\ 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} x_0 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \\ & = \begin{bmatrix} \sum_i x_i^2 & \sum_i x_i \\ \sum_i x_i & n+1 \end{bmatrix} \end{aligned}$$

which matches the other linear regression derivation

Normal Equations

Alternatively, we can get the same result by directly deriving the **normal equations**:

$$Az = y$$

pre-multiply both sides by A^T

$$A^T A z = A^T y$$

$$\Rightarrow z = \underbrace{(A^T A)^{-1}}_{\text{Moore-Penrose pseudo-inverse of } A} A^T y$$

$\hookrightarrow$ Moore-Penrose pseudo-inverse of A

$$A^+ \triangleq (A^T A)^{-1} A^T$$

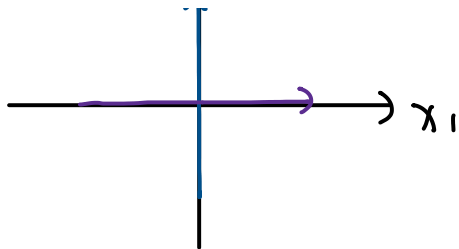
Why Does this Work? A Geometric Interpretation

The fundamental theorem of linear algebra tells us that the null space of A^T is **orthogonal** to the column space of A . For example, consider

$$\uparrow x_2$$

$$A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\dots \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



$$Ax_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A^T x = 0$$

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$x_1 = 0$$

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A Geometric Interpretation Continued

So when we pre-multiply $r = Az - y$ by A^T , we are preserving information in the column space of A (because none of it lies in the kernel of A^T). For linear regression, the column space of

$$A = \begin{bmatrix} x_0 & 1 \\ x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix}$$

is the subspace of $\mathbb{R}^{n+1}$ spanned by our basis functions ($\phi_1(x) = x$ and $\phi_2(x) = 1$) evaluated at our data points.

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解释一下normal equation:

✿ 一、背景：从线性方程组出发

在前面的推导中，我们知道最小二乘问题是：

$$\min_z \|Az - y\|^2$$

其中：

- A ：是特征矩阵（例如，第一列是 x_i ，第二列是全 1）
- z ：是未知参数向量（例如 $z = [a, b]^T$ ）
- y ：是观测数据列向量

💡 二、最小二乘问题的目标

我们想要找到一个 z ，让 Az 尽量接近 y 。

但一般情况下， $Az = y$ **可能无解**（因为 A 可能不是方阵，或方程超定）。
所以我们要求一个**最优近似解**，让误差平方最小。

📘 三、求解思路

我们要求的是：

$$\min_z \|Az - y\|^2 = (Az - y)^T (Az - y)$$

展开并对 z 求导（上一页你已经看到）：

$$\nabla_z f(z) = 2A^T(Az - y) = 0$$

于是得到：

$$A^T Az = A^T y$$

这就是所谓的 **Normal Equation（正规方程）**。

✿ 五、解释紫色部分

Moore–Penrose pseudo-inverse of A

$$A^+ = (A^T A)^{-1} A^T$$

👉 这表示一种**广义逆矩阵（pseudo-inverse）**：

- 当 A 是满秩矩阵时， $(A^T A)^{-1} A^T$ 就是它的**左逆**；
- 当 A 不是方阵（比如样本比参数多）时，这个形式仍然能给出最小二乘意义下的“最优解”。

因此：

$$z = A^+ y$$

所以简单来说，假设我们只处理二维情况的时候，即 Φ 为1和 x_i ，那么我们使用之前的结论；处理更通用的情况（多元情况），我们用pseudo inverse的方法。

这种情况可以处理overdetermined system，这种情况下没有代数解，只能找最佳近似。